METRIC

DOD-E-24618(SH)
18 April 1983

MILITARY SPECIFICATION

ELECTRODES, SILVER-SILVER CHLORIDE, REFERENCE TYPE PORTABLE CELLS AND REPLACEMENT ELEMENTS (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for three types of silver-silver chloride reference electrodes used as components of cathodic protection systems.

1.2 Classification. The reference electrodes shall be furnished in the following types, as specified (see 6.2):

Type I - Silver-silver chloride portable reference electrode (cell).
Type II - Silver-silver chloride power supply reference electrode.
Type III - Silver-silver chloride replacement element kit for MIL-E-23919 reference electrode.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-1035 - Plastic Molding Material, Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes: Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-S-6872 - Soldering Process, General Specification for.
MIL-S-8660 - Silicone Compound.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-W-16878 - Wire, Electrical, Insulated, General Specification for.
MIL-W-16878/12 - Wire, Electrical Fluorinated Ethylene Propylene (FEP) Insulated, 200°C, 1000 Volts.
MIL-I-17695 - Insulation Tape, Electrical, Filler Type, Flameproof, Synthetic.
MIL-I-22076 - Insulation Tubing, Electrical, Nonrigid, Vinyl, Very Low Temperature Grade.
MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-I-23053 - Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for.
MIL-E-23919 - Electrodes, Reference, Circular, Corrosion Preventive, (Flush Type): Silver-Silver Chloride.
MIL-P-25732 - Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Limited Service at 275°F (135°C).

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

AIR FORCE-NAVY AERONAUTICAL

AN6227 - Packing, O-Ring Hydraulic.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be of the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, NW, Washington, DC 20036.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Type I, silver-silver chloride portable reference electrode (cell).

3.1.1 Description. The reference electrode shall consist of an electrode element of treated silver gauze contained in a plastic holder. A silver wire shall connect the electrode element to a silver plated cable which shall be at least 23 meters (75 feet) long. The cable shall extend from a watertight connection with the silver wire and shall be the means by which the electrode is connected to other components of the cathodic protection system (see figure 1).

3.1.2 Material.

3.1.2.1 Silver-silver chloride electrode element. The silver-silver chloride electrode element shall be a 19 millimeters (mm) ± 0.8 mm by 76 mm ± 0.8 mm (3/4 ± 1/32 inch by 3 ± 1/32 inch) piece of silver-chloride treated silver gauze, woven of silver wire 0.25 ± 0.04 mm (10 ± 1.5 mils) in diameter to a no. 40 mesh. Silver wire used in fabricating the gauze shall contain not less than 99.95 percent pure silver.

3.1.2.2 Plastic holder. The plastic holder shall be transparent polyvinyl chloride or polyvinyl chloride-acetate conforming to MIL-I-22076. The nominal inside diameter (i.d.) shall be 9.5 mm (3/8 inch) and the wall thickness shall be 1.6 ± 0.8 mm (1/16 ± 1/32 inch). The plastic holder shall be long enough to enclose the element with a 3.2 ± 0.8 mm (1/8 ± 1/32 inch) extension below the element, the watertight connection, and allow for attachment to the cable. The holder shall have two ventilation holes, 180 ± 10 degrees apart, each with a nominal diameter of 3.2 mm (1/8 inch), approximately 13 mm (1/2 inch) from the top (see figure 1).

3.1.2.3 Watertight connection. The cable to silver wire connection of the electrode shall be watertight. The watertight connection shall consist of 9.5 mm (3/8 inch) i.d. plastic water resistant, heat shrinkable tubing conforming to MIL-I-23053 covering a single layer of insulation tape conforming to MIL-I-17695. The tubing shall be covered by the plastic holder, specified in 3.1.2.2, which shall be secured with a piece of 13 mm (1/2 inch) i.d. heat shrinkable tubing (see figure 1).

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3.1.2.4 Silver wire. The silver wire connecting the electrode to the cable shall be 1.6 mm (1/16 inch) or greater in diameter and at least 50 mm (2 inches) long and contain not less than 99.95 percent pure silver.

3.1.2.5 Cable. Cable shall be size KK-16, silver coated conductor with insulation conforming to MIL-W-16878 and MIL-W-16878/12 and shall be at least 23 meters (75 feet) in length.

3.1.3 Construction.

3.1.3.1 Silver-silver chloride electrode element. The silver gauze shall be rolled in a circular pattern (along the 19 mm (3/4 inch) axis) to fit into the plastic holder with a clearance of approximately 1.5 mm (1/16 inch) around the element. A silver wire, specified in 3.1.2.4, shall be heat swaged along the 19 mm (3/4 inch) edge of the gauze using no solder or flux. The element shall then be treated as follows before insertion in the plastic holder. The gauze shall be cleaned by holding the gauze in a reducing flame of a hydrogen gas burner for approximately 30 seconds (the silver surface should be bright and shiny). The gauze shall be handled with a stainless steel tong after cleaning. The gauze shall be removed from the flame and immediately dipped into molten chemically pure, (CP grade - commercial analyzed reagent) silver chloride maintained at slightly above 455 degrees Celsius (°C). Hold for a sufficient amount of time (approximately 30 seconds) for the gauze to reach the temperature of the silver chloride and to result in a weight gain of at least 1 gram. The gauze shall be allowed to cool for at least 60 seconds, then it shall be returned to the reducing hydrogen flame for 5 to 10 seconds. If the reference potential criteria of 4.3.3 is not met, reheat the gauze in a reducing hydrogen flame for 30 seconds, redip into the molten chemically pure silver chloride for approximately 30 seconds to result in a weight gain of at least 1 gram, then return to the hydrogen flame for 5 to 10 seconds.

3.1.3.2 Watertight connection and holder. The silver wire protruding from the element shall be crimped and silver soldered to the cable. Soldering shall conform to MIL-S-6872. The connection shall be wrapped as specified in 3.1.2.3. The plastic holder shall be fitted over the element and watertight connection and the ventilation holes shall be free from blockage.

3.2 Type II, silver-silver chloride power supply reference electrode.

3.2.1 Description. The reference electrode shall consist of an electrode element of a 65 square centimeter (cm^2) (10 square inch (in^2)) treated silver gauze contained in a plastic holder. A silver wire shall connect the electrode element to a coated KK-14, copper cable conforming to MIL-W-16878 and MIL-W-16878/12 at least 23 meters (75 feet) long. The cable shall extend from the watertight connection with the silver wire and shall be the means by which the electrode is connected to other components of the cathodic protection system (see figure 2).

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3.2.2 Material.

3.2.2.1 Silver-silver chloride electrode element. The silver-silver chloride electrode element shall be a 5 cm ± 0.8 mm by 13 cm ± 0.8 mm (2 ± 1/32 inch by 5 ± 1/32 inch) piece of silver chloride treated silver gauze, woven of silver wire 0.25 ± 0.03 mm (10 ± 1.5 mils) in diameter to a no. 40 mesh. The silver wire used in fabricating the gauze shall contain not less than 99.95 percent pure silver.

3.2.2.2 Plastic holder. The plastic holder shall be transparent polyvinyl chloride or polyvinyl chloride-acetate conforming to MIL-I-22076. The holder shall be a 6.4 cm ± 3 mm (2-1/2 ± 1/8 inch) cylinder covering the length of the element and necking down to an i.d. of 9.5 ± 0.8 mm (3/8 ± 1/32 inch) to allow attachment to the cable. The wall thickness shall be 3.2 ± 0.8 mm (1/8 ± 1/32 inch). The holder shall have four ventilation holes, 90 ± 10 degrees apart, each with a nominal diameter of 9.5 mm (3/8 inch), located in the shoulder area. The holder shall be fitted with a screen bottom. The screen shall be polyvinyl chloride composition A, type II, grade GU, conforming to L-P-1035. The screen shall have twelve uniformly spaced holes, each with a diameter of 1.6 mm (1/16 inch).

3.2.2.3 Watertight connection. The cable to silver wire connection of the electrode shall be watertight. The watertight connection shall consist of 9.5 mm (3/8 inch) i.d. plastic water resistant, heat shrinkable tubing conforming to MIL-I-23053 covering a single layer of insulation tape conforming to MIL-I-17695. The tubing shall be covered by the plastic holder, specified in 3.2.2.2, which shall be secured with a piece of 19 mm (3/4 inch) i.d. heat shrinkable tubing (see figure 2).

3.2.2.4 Silver wire. The silver wire connecting the electrode element to the cable shall contain not less than 99.95 percent pure silver. The silver wire shall be at least 1.6 mm (1/16 inch) in diameter and 7.6 cm (3 inches) long.

3.2.2.5 Cable. Cable shall be of size KK-14, tinned wire conductor with insulation conforming to MIL-W-16878 and MIL-W-16878/12 and shall be at least 23 meters (75 feet) in length.

3.2.3 Construction.

3.2.3.1 Silver-silver chloride electrode element. The silver gauze shall be rolled in a circular pattern (along the 5 cm (2 inches) axis) to fit into the plastic holder with a clearance of approximately 3.2 mm (1/8 inch) around the element. A silver wire, specified in 3.1.2.4, shall be heat swaged along the 5 cm (2 inches) edge of the gauze using no solder or flux. The element shall then be treated as follows before insertion in the plastic holder. The gauze shall be cleaned by holding the gauze in a reducing flame of a hydrogen gas burner for approximately 30 seconds (the silver surface should be bright and shiny). The gauze shall be handled with a stainless steel tong after cleaning. The gauze shall be removed from the flame and immediately dipped into the molten chemically pure, (CP grade - commercial analyzed reagent) silver chloride maintained at slightly above 455°C. Hold for a sufficient amount of time (approximately 30 seconds) for the gauze to reach the temperature of the silver

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chloride and to result in a deposition of at least 5 grams. The gauze shall be allowed to cool for approximately 60 seconds, then it shall be returned to the reducing hydrogen flame for 5 to 10 seconds. If the reference potential criteria of 4.3.3 is not met, reheat the gauze in a reducing hydrogen flame for 30 seconds and redip into the molten chemically pure silver chloride for approximately 30 seconds to result in a weight gain of at least 5 grams (0.18 ounce).

3.2.3.2 Plastic holder and screen. The plastic holder and screen may be molded or machined as separate pieces.

3.2.3.3 Watertight connection and holder. The silver wire protruding from the element shall be crimped and silver soldered to the cable. Soldering shall conform to MIL-S-6872. The connection shall be wrapped as specified in 3.2.2.3. The plastic holder shall be fitted over the element and watertight connection and the ventilation holes shall be free from blockage. The screen shall be cemented to the holder using copolymer conforming to 3.3.2.2.

3.3 Type III, silver-silver chloride replacement element kit.

3.3.1 Description. The replacement element shall be a 400 cm^2 (64 in^2) piece of treated silver gauze enclosed in a copolymer screen pouch with at least an 18 cm (7 inches) protruding silver wire for the electrical connection during element replacement. Two O-rings and a 7 grams (0.25 ounce) container of silicone grease shall be included with the element (see figure 3).

3.3.2 Material.

3.3.2.1 Silver-silver chloride electrode element. The silver-silver chloride electrode element shall be a 40.6 cm $\pm$ 0.8 mm by 10.2 cm $\pm$ 0.8 mm (16 $\pm$ 1/32 inch by 4 $\pm$ 1/32 inch) piece of silver chloride treated silver gauze, woven of silver wire 0.25 $\pm$ 0.04 mm (10 $\pm$ 1.5 mils) in diameter to a no. 40 mesh. The silver wire used in fabricating the gauze shall contain not less than 99.95 percent pure silver.

3.3.2.2 Vinylidene chloride/vinyl chloride copolymer cloth. The gauze shall be wrapped in a 2.5 cm (1 inch) wide fine filament screen woven of copolymer wire (copolymer of vinylidene chloride and vinyl chloride) 0.4 $\pm$ 0.04 mm (15 $\pm$ 1.5 mils) in diameter, and shall be enclosed in a pouch-type bag with drawstring of filament wire 35.5 cm long $\pm$ 2.5 cm (14 inches long $\pm$ 1 inch). The bag shall be made of copolymer screen, woven of vinylidene chloride/vinyl chloride wire 0.4 $\pm$ 0.04 mm (15 $\pm$ 1.5 mils) in diameter.

3.3.2.3 Silver wire. The silver wire shall be 2.4 $\pm$ 0.4 mm (3/32 $\pm$ 1/64 inch) in diameter and at least 23 cm (9 inches) long and shall contain not less than 99.95 percent pure silver. The silver wire shall extend from the bag described in 3.3.2.2 at least 18 cm (7 inches) to allow attachment to the projecting contact pin (see figure 3).

3.3.2.4 O-rings. The two O-rings for the hub of the base assembly, as specified in MIL-E-23919, (see figure 3) shall be circular rubber rings conforming to MIL-P-25732. Size and shape shall be AN6227-13 as specified in AN6227. Sealing grease for installation of the O-rings shall conform to MIL-S-8660.

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3.3.3 Silver-silver chloride electrode element construction. The silver gauze shall be folded with an accordion pleat to form a 10 cm ± 0.4 mm (4 ± 1/32 inch) square. A silver wire (2.4 ± 0.4 mm (3/32 ± 1/64 inch) diameter) shall be swaged in the center of the first edge of the gauze using no solder or flux. The element shall then be treated as specified in 3.1.3.1 (except the weight gain should be approximately 30 grams). After the element has cooled the corners shall be trimmed to 11 cm ± 0.4 mm (4-3/8 ± 1/32 inch) diameter using stainless steel shears (see figure 3). The element shall be prepared as specified in 3.3.2.2 before insertion in the plastic holder described in MIL-E-23919.

3.4 Workmanship.

3.4.1 Plastic holder. The plastic holder shall be smooth and free of burns, inclusions, bubbles, and similar defects.

3.4.2 Watertight connection. Heat shrinkable tubing shall be free of splits, cracks, burn spots, and other similar defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Visual examination. Each electrode and component shall be visually examined to assure compliance with 3.4.1 and 3.4.2. External dimensions of the holders, cabling and connections shall be checked to assure compliance with appropriate areas of the section. Failure of the electrode or components to meet the criteria of section 3 shall be cause for rejection.

4.3 Tests. Each electrode shall be tested as specified in 4.3.1 through 4.3.3.

4.3.1 Pull test. Each electrode shall have the gauze to silver wire and the wire to cable connection tested for a 4.5 kilogram (10 pound) tension load after complete assembly. If any connection fails, the electrode shall be rejected. Electrodes which fail this test shall not be repaired or accepted for retest.

4.3.2 Electrical continuity. Each electrode shall be tested for electrical continuity with an ohm meter or similar resistance measuring device after the pull test is made. One terminal of the tester shall be connected to the electrical lead and the other terminal shall contact the active electrode surface. A steady reading of resistance of greater than 2 ohms shall constitute failure and shall be cause for rejection of the electrode.

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4.3.3 Reference comparison test. Each electrode shall be tested against a standard reference saturated calomel electrode which can be traced to the U.S. Bureau of Standards standard reference. Electrodes shall be tested in 25 ± 1 ohm-centimeter, 3.5 percent sodium chloride solution at 28°C. Electrodes shall be allowed to stabilize (open circuit) in the sodium chloride solution approximately 15 minutes before testing. The comparison tests shall be made with a volt meter with an accuracy to plus or minus 0.1 millivolts (mV) and an input impedance of at least 1 megohm. Deviation voltage by the test cell from the standard by more than 10 mV shall be cause for rejection of electrodes. Electrodes rejected may be resubmitted for retest after rework by heating the gauze with a hydrogen torch, redipping and returning to the hydrogen flame (see 3.1.3.1).

4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I and type II electrodes. Type I and type II electrodes shall be individually protected in accordance with MIL-P-116, method IC, and the following:

(a) Connecting cables shall be coiled to a minimum diameter that will not kink or place undue stress on the cable. The cable shall be secured in two or more places with plastic, cloth or wire ties to prevent unravelling.
(b) Each electrode shall be placed in a plastic bag fabricated from material conforming to MIL-B-22191, type I. Bag closure may be by heat sealing with enough material that two additional heat seals may be effected for inspection purposes. Alternatively, the bag closure may be of a recloseable type that will assure the requirements for method IC.

5.1.1.2 Type III, replacement element. Replacement elements should be individually packaged in plastic bags as specified in 5.1.1.1(b).

5.1.1.3 Unit containers. Packaged electrodes and elements shall be individually unit protected in a paperboard folding, set-up or metal stayed box or a fiberboard box conforming to PPP-B-566, PPP-B-676, PPP-B-665 or PPP-B-636, respectively, with type, class or style at the suppliers option. Box closure shall conform to the applicable container specification or appendix thereto. Closure methods I and V shall apply to domestic and weather resistant containers, respectively, for PPP-B-636 boxes.

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5.1.2 Level C. Preservation-packaging of electrodes and replacement elements shall afford protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. As a minimum, each reference electrode shall be packaged in a reclosable air tight plastic bag. The contractor's normal retail or wholesale preservation-packaging methods may be utilized when such meet the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Electrodes and elements packaged as specified in 5.1 shall be packed in containers conforming to any one of the following specifications, with container selection at the option of the contractor.

Specification	Container	Type or class
PPP-B-601	Box, Wood, Cleated-Plywood	Overseas type
PPP-B-621	Box, Wood, Nailed and Lock-Corner	Overseas, class
PPP-B-636	Boxes, Shipping, Fiberboard	Weather resistant
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall	Class 2

5.2.1.1 Closure, case liners and gross weight. Shipping containers shall be closed, strapped or banded in accordance with the applicable container specification or appendix thereto, with method V closure applicable to PPP-B-636 boxes. Reinforcing for fiberboard boxes shall be accomplished by use of non-metallic banding or pressure sensitive reinforced tape at the contractor's option. Unless otherwise specified (see 6.2), shipping containers shall have case liners conforming to MIL-L-10547. Case liners shall be closed and sealed in accordance with MIL-L-10547. Case liners are not required for PPP-B-636 fiberboard containers with method V closure nor PPP-B-640 fiberboard containers when sealed with tape as specified in the appendix to PPP-B-640. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification. Wood or wood cleated plywood boxes shall not exceed 200 pounds.

5.2.2 Level B. Electrodes and elements packaged as specified in 5.1 shall be packed in containers as specified in 5.2.1 except that the containers shall be of the domestic type or class. Case liners are not required, and method I closure shall apply for PPP-B-636 boxes.

5.2.3 Level C. Electrodes and elements packaged as specified in 5.1 shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers packing or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2) interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Reference electrodes, components of cathodic protection systems, covered by this specification are intended for the uses specified in 6.1.1 through 6.1.3.

6.1.1 Type I. Type I reference electrode is intended for hull potential measurements with portable volt meters and voltage recorders with input impedance greater than 20,000 ohms.

6.1.2 Type II. Type II reference electrode is intended for use with automatic control impressed current cathodic protection systems (see MIL-P-15736/9) power supply for underwater surfaces of inactive ships.

6.1.3 Type III. Type III reference electrode is intended for element replacement of reference electrodes (see MIL-E-23919) for impressed current systems on active ships.

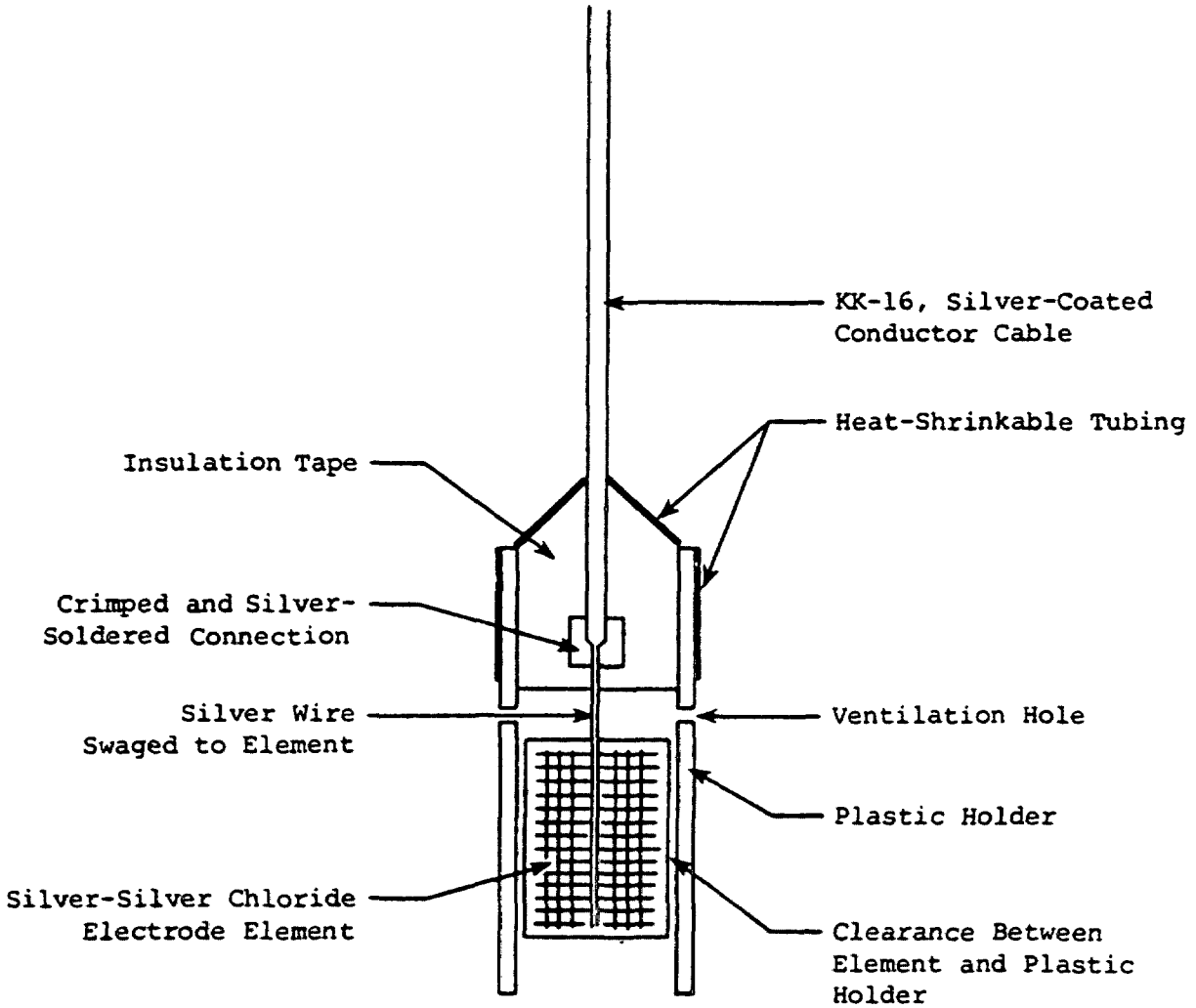
6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of reference electrode (see 1.2).
(c) Packaging, packing or marking requirements (see 5.3).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 2040-N145)

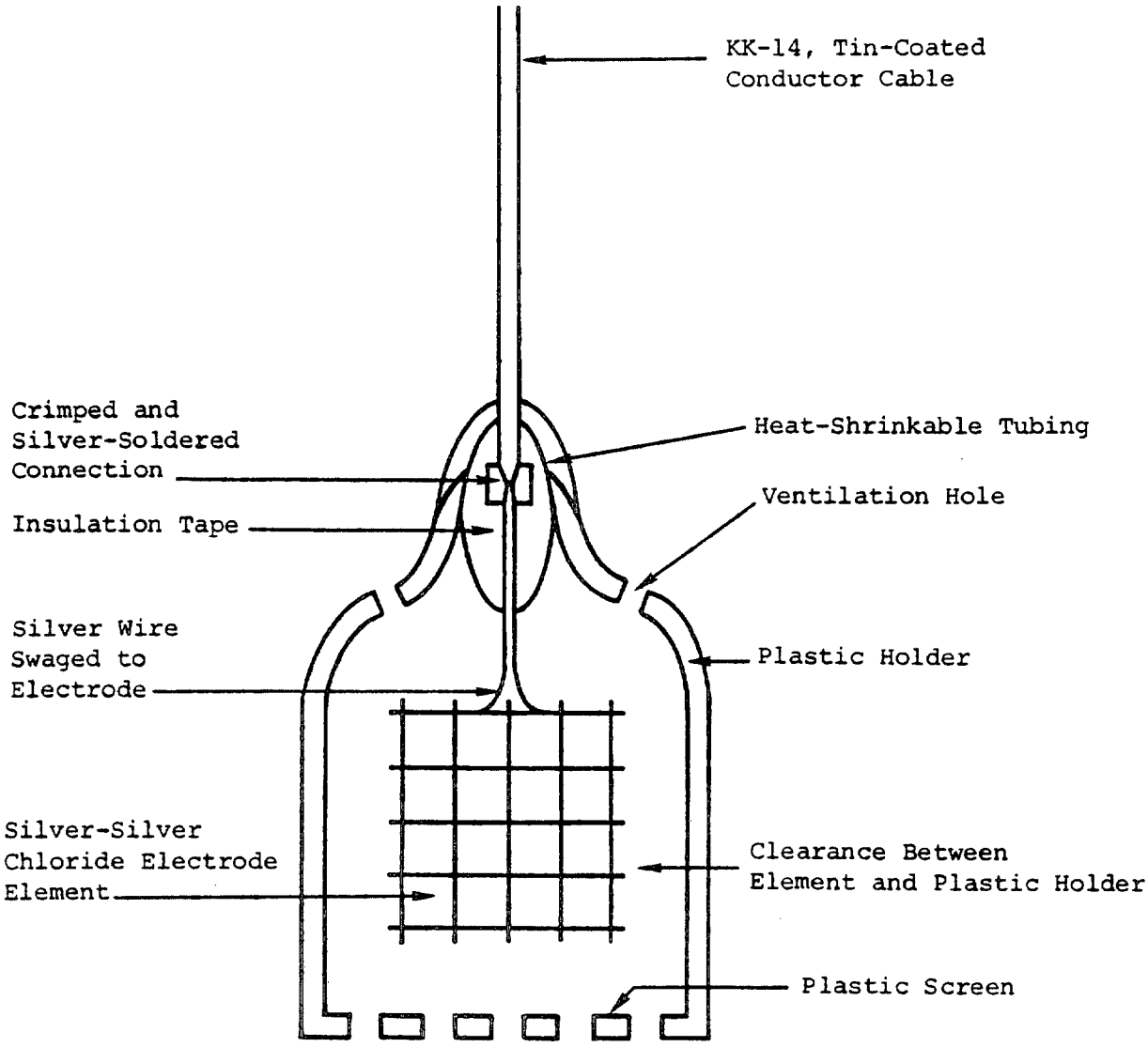
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SH 12207

FIGURE 1. Cross section of type I reference electrode, general configuration.

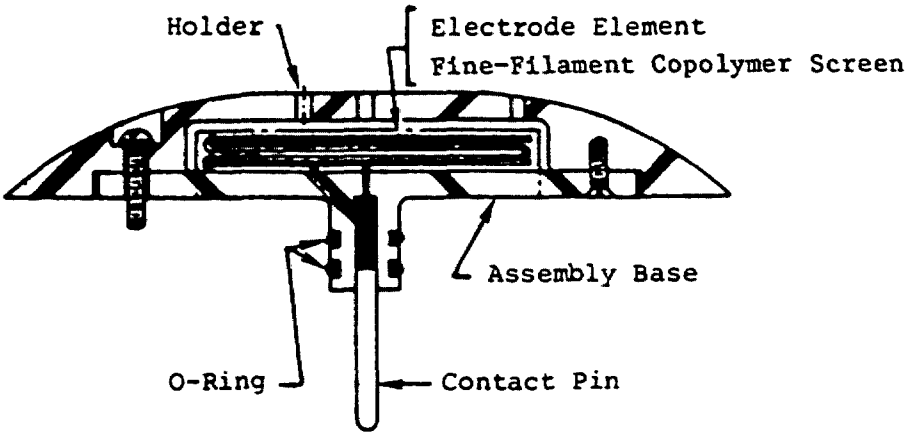
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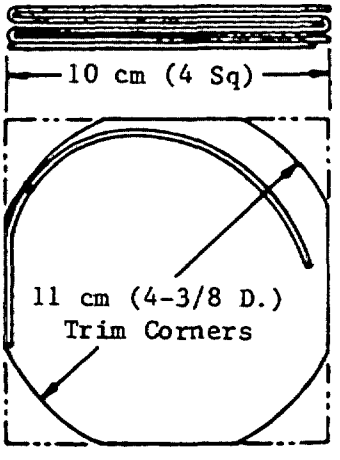
SH 12208

FIGURE 2. Cross section of type II reference electrode, general configuration.

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MIL-E-23919 Reference Electrode, Assembled



Electrode Element

Dimensions are in centimeters (inches)
Tolerance: ± 0.04 cm (± 1/32) unless specified

NOTE: Two O-rings and silicone grease package not shown.

SH 12209

FIGURE 3. Reference electrode element, silver-silver chloride, type III.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-E-24618(SH)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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